INJECTORS

1. Use two injectors of equal capacity, each capable of supplying the percentage of the maximum cylinder steam consumption, corresponding to the given fuel and service, as per table below:

CYLINDERS	STEAM	MAXIMUM STEAM CONSUMPTION PER CYLINDER H. P. HOUR	
		POUNDS	GALLONS
Simple	Saturated	27.0	3.24
	Superheated	20.8	2.50
Compound	Saturated	23.5	2.82
	Superheated	19.7	2.36

FUEL	SERVICE	% MAXIMUM STEAM CONSUMPTION SUPPLIED BY ONE INJECTOR
Other than Oil or Pulverized Coal	Road	75%
	Switching	60%
Oil or Pulverized Coal	Road	87%
	Switching	70%

This method assumes that the boiler and cylinder horse power are equal (a 100% engine). When boiler horse power is less than 90% or over 100% of cylinder horse power, make proportionate decrease or increase in capacity of injectors.

TABLE OF INJECTOR CAPACITIES AT 200 LB. BOILER PRESSURE

SIZE NO.	NATHAN			SELLERS	OHIO		HANCOCK	AVERAGE CAPACITIES GALLONS PER HOUR
	MONITOR & ALL N. L. EXCEPT SIMPLEX	SIMPLEX LIFTING & N. L. EXCEPT N. W.	SIMPLEX N. L. N. W.	ALL TYPES	STD'S A. B. & E.	STD'S C. & D.	TYPES A. B. & D.	
7	1920	1950	1950		1800		2040	1932
7½				2010				2010
8	2340	2580	2580		2300	2700	2710	2525
8½				2587				2587
9	2940	3240	3240		2900	3200	2980	3083
9½				3187	3150		3300	3212
10	3750	3800	3800		3400	3800	4060	3768
10½				3952	3600			3776
11	4150	4550	4550		3800	4200	4420	4278
11½				4725				4725
12	4450	5200	5500		4200	4600		4790
12¾0				5700				5700
13		6000	6000–7000					6250
13¾0				6800				6840
14		7500						7500

For pressures less than 200 lb. use percentages of the capacities in table at left as follows:

BOILER PRESSURE	PER CENT
150	92
160	94
170	96
180	98
190	100

2. Lifting injectors and inspirators must be so located that bottom of overflow nozzle will be within the limits of 2" to 6" above the highest water line of tank. Non-lifting inspirator operating valves must be located above the highest water line of tank.

3. All injectors, whether equipped with open or closed overflow must have overflow pipe securely braced at two or more points along its course between injector and its termination, so that pipe will not be jarred loose from injector.

   Lifting Injectors located ahead of engineer, and non-lifting injectors having overflow in cab, must have closed over-flow connections when applied to engines built for operation in the United States. Material orders must specify: "closed over-flow pipe connection."

   Overflow from non-lifting injectors must not discharge into ash pan, nor at any other point above injectors.

4. All classes of engines to have copper steam Pipes and steel suction and over-flow pipes. Use steel delivery pipes outside of boiler, and copper inside of boiler.

   When injector checks are attached to backhead of boiler, it is very important that the forward end of the delivery pipe be bent down and always covered with at least six inches of water. Locate check as low as possible.

   Locate delivery pipes leading to top-inlet checks so that they will be self-draining to their lowest point.

5. Use Line Checks only when specified by R. R. Co., or with injectors where makers furnish same as part of equipment.

QUESTION. Duplicate engines.
EXCEPTIONS: R. R. Co.'s drawings, specifications or instructions to contrary.